

Python Strings - 30 Moderately Challenging Questions

Interview & Practice Worksheet | Output prediction • Edge cases • Short code snippets

1. Indexing vs Slicing

What is the difference between indexing and slicing in Python strings? Explain what happens with a negative index and give one example of each.

2. Output Prediction

Predict the output:

```
s = "Python"
print(s[1:5])
print(s[-4:-1])
```

3. lower() vs casefold()

What is the difference between lower() and casefold()? Why can casefold() be more suitable for case-insensitive comparisons?

4. Output Prediction

Predict the output:

```
a = "Python"
b = "python"
print(a.lower() == b)
print(a.casefold() == b.casefold())
```

5. == vs is

What is the difference between == and is when comparing strings? Why should is generally not be used to compare string values?

6. Output + Concept

Predict the output and explain:

```
a = "hello"
b = "hello"
print(a == b)
print(a is b)
```

Would you consider the result of is guaranteed for equal string values?

7. encode() vs decode()

What is the difference between encode() and decode()? What types do they accept and return?

8. Output Prediction

Predict the output:

```
text = "Python"
data = text.encode("utf-8")
print(type(data))
print(data.decode("utf-8"))
```

9. Raw Strings

What is a raw string? Explain why r"C:\new\test" behaves differently from "C:\new\test".

10. Escape Sequences

What is the output?

```
s = r"\n\tPython"
print(s)
print(len(s))
```

Why does len(s) not count a newline and tab here?

11. String Immutability

What are escape sequences in Python strings? Explain the purpose of \n, \t, \\, \', and \".

12. Slicing Edge Cases

What happens here?

```
s = "Python"
s[0] = "J"
print(s)
```

Explain why the statement fails and how you would create "Jython" instead.

13. join() vs +

Predict the output:

```
s = "Python"
print(s[0:100])
print(s[-100:100])
print(s[100:200])
```

14. join() vs str()

Why is join() generally preferred over repeated + when joining many strings? Mention both readability and performance.

15. Immutable Methods

Predict the exact output:

```
print("".join(["P", "y", "t", "h", "o", "n"]))
print(str(["P", "y", "t", "h", "o", "n"]))
```

16. split() vs partition()

Why are Python strings immutable? How does immutability affect operations such as replace(), upper(), and slicing?

17. strip() Edge Case

Predict the output:

```
s = "hello"
result = s.upper()
print(s)
print(result)
```

18. find() vs index()

What is the difference between split() and partition()? Consider the number and type of values returned by each.

19. find() Output

Predict the output:

```
s = "one,two,three"  
print(s.split(","))  
print(s.partition(","))
```

20. startswith() / endswith()

What does this return?

```
print(" Python ".strip())  
print("###Python###".strip("#"))
```

Why does strip() not remove characters from the middle of a string?

21. String Validation

What is the difference between find() and index()? What happens when the searched substring is not present?

22. isdigit() / isalnum()

Predict the output:

```
s = "banana"  
print(s.find("na"))  
print(s.index("na"))  
print(s.find("xyz"))
```

What would happen if you replaced find("xyz") with index("xyz")?

23. capitalize() / title() / upper()

What is the difference between startswith() and endswith()? How can you use them with a tuple of possible prefixes or suffixes?

24. Case Conversion Output

Predict the output:

```
filename = "report.PDF"  
print(filename.endswith(".pdf"))  
print(filename.lower().endswith(".pdf"))
```

25. Negative Indexing

What is the difference between isalpha(), isdigit(), and isalnum()? Give one example where isalnum() returns True but isalpha() returns False.

26. Negative Index Edge Case

Predict the output:

```
print("123".isdigit())  
print("12.3".isdigit())  
print("Python3".isalnum())  
print("Python 3".isalnum())
```

27. String Quotes

What is the difference between capitalize(), title(), and upper()? Explain how they behave on the string "python programming".

28. String Multiplication

Predict the output:

```
s = "python programming"
print(s.capitalize())
print(s.title())
print(s.upper())
```

29. in Operator

How does negative indexing work in strings? What happens if you use an index whose absolute value is larger than the string length?

30. str() vs repr()

What is the difference between str() and repr() when displaying a string? Why can repr() be useful while debugging?

Predict the output:

```
s = "Hello\nPython"
print(str(s))
print(repr(s))
```

31. Reverse with Slicing

How can you reverse a string using slicing? Explain the meaning of the step value -1.

Predict the output:

```
s = "Python"
print(s[::-1])
print(s[4:1:-1])
```

32. F-Strings

What are f-strings? Predict the output:

```
name = "Nayan"
age = 21
print(f'{name} is {age} years old.')
print(f'Next year: {age + 1}')
Why is the expression inside {} evaluated?
```

33. split() vs splitlines()

What is the difference between split() and splitlines()? Predict the output:

```
text = "Python\nJava\nC++"
print(text.split())
print(text.splitlines())
```

34. Unicode + Encoding

What is Unicode, and why does it matter for Python strings?

Predict the output:

```
text = "café"
data = text.encode("utf-8")
print(len(text))
print(len(data))
print(data.decode("utf-8"))
Why can len(text) and len(data) be different?
```